Rotation

How to rotate a vector through an angle θ (counter-clockwise)? Probably the simplest way is to memorize the matrix for rotation, and then test it by rotating unit vectors through 90°. The matrix is

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Test it

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

That looks good. So

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

means that

$$u = x\cos\theta - y\sin\theta$$

$$v = x\sin\theta + y\cos\theta$$

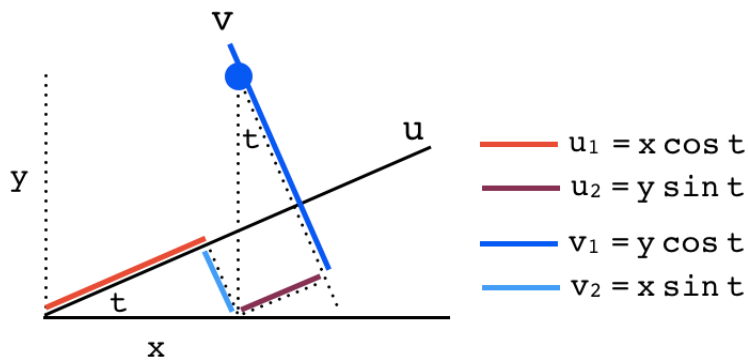
Rotating the points counter-clockwise is equivalent to rotating the coordinate system clockwise.

rotation by $+\theta$ then $-\theta$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

standard derivation

Here is a second derivation with a sort of minimalist diagram of rotation



We draw the horizontal x-axis and the rotated u-axis. The angle between them is t. We plot our point and then draw perpendiculars to both axes. To finish the set-up, we draw perpendiculars from the point $(x, 0)$ as shown.

Once the drawing is rendered, we are almost done. Now we just work our way through.

$$u = u_1 + u_2 = x \cos t + y \sin t$$

All the triangles in the diagram are similar, with small angle t.

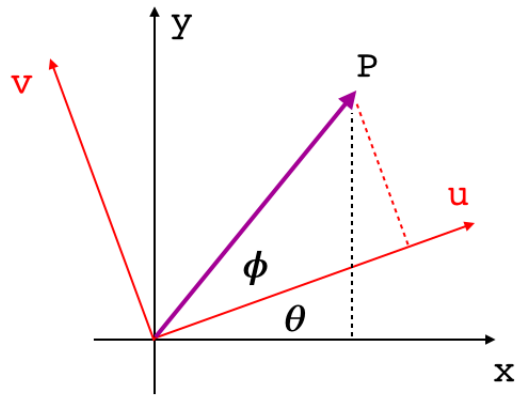
$$v = -v_2 + v_1 = -x \sin t + y \cos t$$

I find this hard to remember. You will know that you've done it right if you have both x and y as the *hypotenuse of a right triangle*. Start with the one for x and then rotate the triangle the other direction about the point $(x, 0)$. Stretch to fit.

Here we are rotating the coordinate system counter-clockwise. Hence, the minus sign is in the formula for the *vertical* component, v.

Stewart

I found a nice method in Stewart that depends on knowing the sum of angles formulas, which I urge you to memorize.



Here, we have the standard xy-coordinates in black. The rotated uv-coordinate system is in red, with a rotation angle θ in the counter-clockwise direction.

The ray to the point P has length r. The coordinates of point P in the u, v system are naturally expressed in terms of ϕ:

$$u = r\cos\phi$$

$$v = r\sin\phi$$

while x and y are naturally expressed in terms of the combined angle $\theta + \phi$.

$$x = r\cos(\theta + \phi)$$

Now, use the sum formula for cosine:

$$x = r\cos\theta\cos\phi - r\sin\theta\sin\phi$$

But we have from above that

$$u = r\cos\phi$$

$$v = r \sin \phi$$

So

$$x = u \cos \theta - v \sin \theta$$

It's as easy as that.

In the same way:

$$\begin{aligned} y &= r \sin(\theta + \phi) \\ &= r \sin \theta \cos \phi + r \cos \theta \sin \phi \\ &= u \sin \theta + v \cos \theta \end{aligned}$$

The only trick here is that we have x, y in terms of u, v.

Solve for u and v

To convert these formulas to functions $u = f(x, y)$ and $v = f(x, y)$, there is a hard way and an easy way. The easy way is to switch x, y for u, v and at the same time, substitute $-\theta$ for θ. What this amounts to is relabeling our diagram with x, y being the rotated axes, and then rotating in the opposite direction (cw rather than ccw).

$$x = u \cos \theta - v \sin \theta$$

switch

$$\begin{aligned} u &= x \cos -\theta - y \sin -\theta \\ &= x \cos \theta + y \sin \theta \end{aligned}$$

(Recall that $\cos -x = \cos x$ and $\sin -x = -\sin x$).

For y

$$y = u \sin \theta + v \cos \theta$$

switch

$$v = x \sin -\theta + y \cos -\theta$$

$$v = -x \sin \theta + y \cos \theta$$

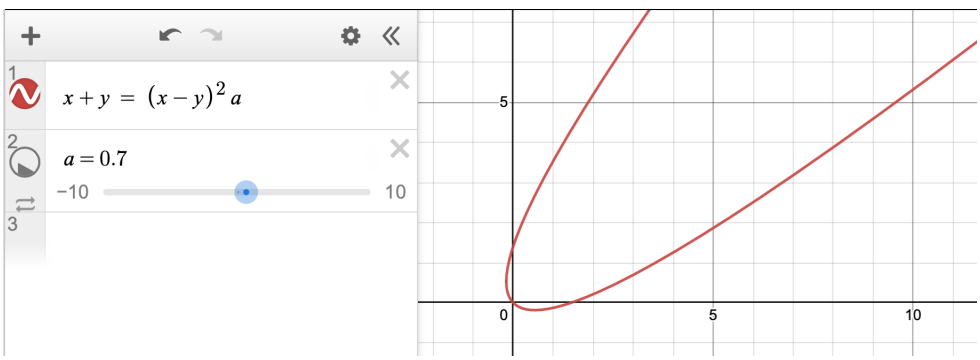
examples

Consider $y = x^2$ and rotate through an angle of 45° cw. The points are rotated cw and the coordinate system ccw, so the minus sign is in the expression for x'.

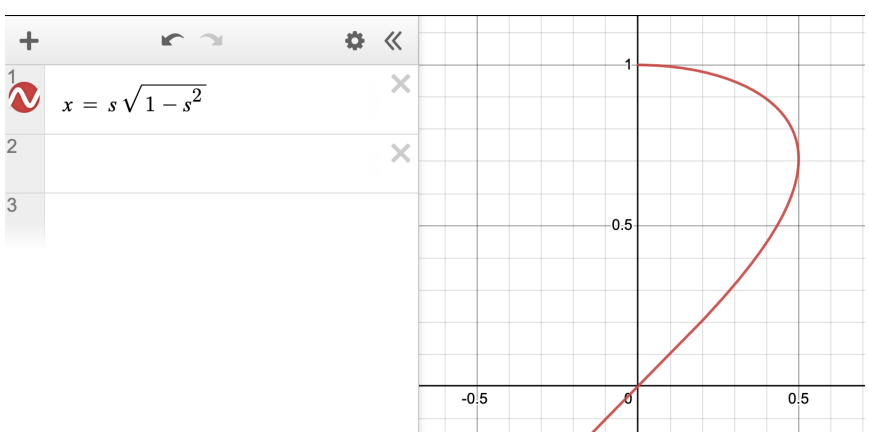
$$x' = 1/\sqrt{2}(x - y)$$
$$y' = 1/\sqrt{2}(x + y)$$

so

$$y' = x'^2$$
$$1/\sqrt{2} \cdot (x + y) = 1/2 \cdot (x - y)^2$$
$$x + y = 1/\sqrt{2} \cdot (x - y)^2$$



Here's another form — both rotated and translated.



Going back to our first result:

$$x' = x\cos\theta - y\sin\theta$$

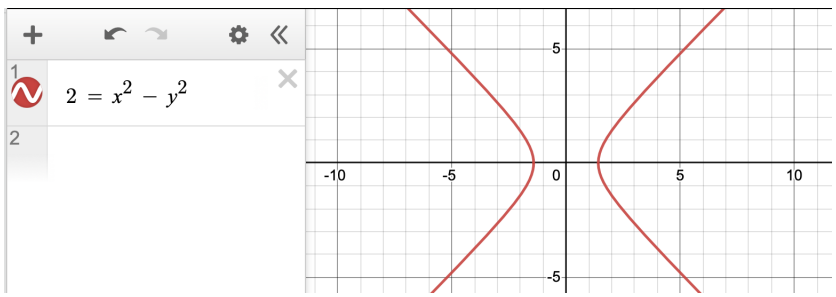
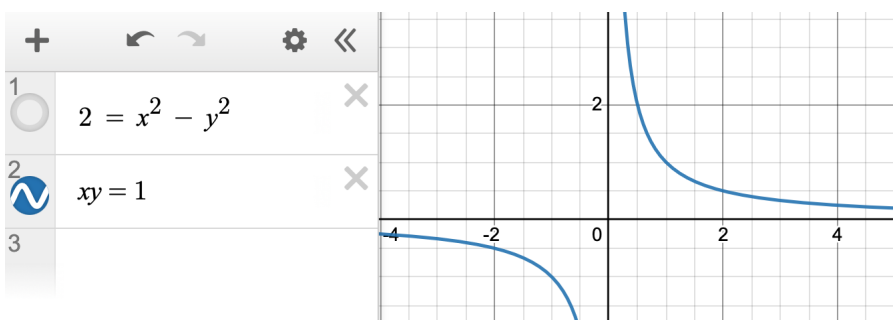
$$y' = x\sin\theta + y\cos\theta$$

Let $x'y' = 1$, that is, the graph is a hyperbola. We turn through an angle of 45° cw.

$$(x\cos\theta - y\sin\theta)(x\sin\theta + y\cos\theta)$$

$$\frac{1}{2}\cdot(x-y)(x+y)$$

$$2 = (x^2 - y^2)$$



When this guy is rotated by 45 ccw, it becomes the familiar graph of $xy = 1$.